

Problem 1:

If the lifetime of the delayed neutrons of group j is:

$$l_{dj} = l_p + \frac{1}{\lambda_j} \quad j = 1, \dots, 6,$$

then the averaged neutron lifetime can be given as:

$$\bar{l} = (1 - \beta)l_p + \sum_{j=1}^6 \beta_j l_{dj} \quad j = 1, \dots, 6,$$

where l_p is the lifetime of the prompt neutrons and λ_j is the decay constant of the delayed neutron group j .

Problem 3:

1: Apply linear interpolation in the group around 1 MeV.

2: For example, the fraction of prompt neutrons that cause fission in ^{238}U is:

$$f_p^{238\text{U}} = \frac{\sum_{E \geq 1\text{MeV}} \bar{\chi}_{p,g} \Delta E_g}{\sum_{g=1}^G \bar{\chi}_{p,g} \Delta E_g}$$

3: Values in the table are given at the group boundaries. Dashes (-) correspond to zero (0).